

EasyClassifier - User Guide

EasyClassifier builds and evaluates classification models from your data file by asking you simple questions. You do not write any code. This guide takes you from an empty computer to your first report.

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1. What you need

Computer	Windows 10 or 11, macOS, or Linux
Python	Version 3.10 or newer . 3.12, 3.13 or 3.14 are recommended (3.10 stops receiving security updates at the end of October 2026). https://www.python.org/downloads/
Disk space	About 400 MB for Python's scientific libraries
Internet	Only while installing
Your data	A .csv file or an Excel .xlsx file (see section 5)
Optional	A LaTeX program, if you want the report as a PDF (section 9)

You will type a few commands into a **terminal** (a window where you type instructions). Every command is given below exactly as you should type it.

- **Windows:** the terminal is called *Command Prompt*. Open it from the Start menu by typing cmd and pressing Enter.
- **macOS:** the terminal is called *Terminal*. Open it with Spotlight (⌘ + Space, type Terminal, press Return).
- **Linux:** open *Terminal* from your applications menu.

2. Install Python

Skip this section if Python 3.10 or newer is already installed. To check, type in the terminal:

Windows	macOS / Linux
py --version	python3 --version

If you see Python 3.10 or a higher number, go to [section 3](#).

Windows

1. Install the **Python install manager**, the tool recommended by python.org for Windows. The easiest way is from the **Microsoft Store**: open the Store, search for *Python install manager*, and click *Get / Install*. (It can also be downloaded from python.org/downloads/windows.)

2. Open **Command Prompt** and type:

```
py --version
```

The first time, the install manager may ask a few configuration questions; the suggested answers are fine. If you then see a line such as `Python 3.14.x`, Python is ready.

3. If instead you are told that no Python is installed, type the line below, then check again with `py --version`:

```
py install 3.14
```

Already using the classic python.org installer? That works too. During installation, tick **“Add python.exe to PATH”**. All the `py` commands in this guide work the same way.

macOS

1. Go to python.org/downloads/macos and download the latest **macOS 64-bit universal2 installer** for Python 3.13 or 3.14.
2. Open the downloaded file and follow the steps.
3. Open **Terminal** and check:

```
python3 --version
```

Linux

Python 3 is usually already installed. On Ubuntu or Debian you also need the tool for separate environments:

```
sudo apt install python3-venv python3-pip
```

3. Install EasyClassifier

Windows

In **Command Prompt**:

```
py -m pip install easyclassifier
```

macOS

In **Terminal**:

```
python3 -m pip install easyclassifier
```

If you installed Python with **Homebrew** and see the message *externally-managed-environment*, follow the Linux steps below instead.

Linux (and Homebrew Python on macOS)

Recent Linux systems ask you to install programs like this one into a separate *environment*. Type these three lines:

```
python3 -m venv ~/easyclassifier-env
~/easyclassifier-env/bin/python -m pip install easyclassifier
~/easyclassifier-env/bin/python -m easyclassifier --version
```

From now on, start EasyClassifier with:

```
~/easyclassifier-env/bin/python -m easyclassifier
```

Check the installation

Windows	macOS	Linux
<code>py -m easyclassifier --version</code>	<code>python3 -m easyclassifier --version</code>	<code>~/easyclassifier-env/bin/python -m easyclassifier --version</code>

You should see something like:

```
EasyClassifier 0.8.2 (Python 3.13.2)
```

Installation downloads EasyClassifier and the scientific libraries it uses (pandas, NumPy, scikit-learn, matplotlib, openpyxl). This takes one to a few minutes.

4. Try it with the demo (2 minutes)

Start EasyClassifier:

Windows	macOS	Linux
<code>py -m easyclassifier</code>	<code>python3 -m easyclassifier</code>	<code>~/easyclassifier-env/bin/python -m easyclassifier</code>

The shorter command `easyclassifier` also works on many computers. If it says *not recognized* or *command not found*, use the commands above; they always work.

Then:

1. Press **ENTER** to choose *Beginner* mode.
2. When asked for your data file, type **demo** and press ENTER.
3. Press **ENTER** at every remaining question to accept the suggested answers.

EasyClassifier analyses a small built-in dataset (150 Iris flowers, 3 species) with seven classifiers and finishes in well under a minute. At the end it tells you where the results are:

```
[OK] All results were saved in this folder:
C:\Users\you\Results\demo_iris_2026-09-28_10-15-02
```

```
...
Finished. Start with report.pdf: it explains the results in plain words
and includes a Methods paragraph you can adapt for a paper.
```

Open that folder and look at `report.pdf` (or `report.tex` if you have no LaTeX, see [section 9](#)).

Useful keys at any question

Type	What it does
ENTER	accept the suggested answer (shown as <i>default</i> or <i>ENTER = ...</i>)
a number	choose that option
? and a number, e.g. ?2	explain option 2 in plain words
Ctrl + C	stop EasyClassifier

5. Prepare your own data

EasyClassifier reads .csv files and Excel .xlsx files (the first sheet). Your table should look like this:

PatientAge	BloodPressure	Smoker	City	Diagnosis
54	130	Yes	Amman	Healthy
61	145	No	Irbid	Sick
47		No	Zarqa	Healthy

Checklist:

- **One row per case** (patient, student, customer, sample ...).
- **The first row holds the column names.**
- **One column holds the groups you want to predict** (here *Diagnosis*). If it contains numbers such as a score or an income, EasyClassifier will offer to split them into groups (Low / Medium / High, or at a value you choose).
- **Empty cells are fine**; EasyClassifier fills them in or removes those rows – you choose.
- **Numbers must be plain numbers**: 72.5, not 72.5 kg. Decimal commas (72,5) are fine in files saved by Excel with semicolons.
- **No totals, notes, merged cells or charts** in the sheet – just the table.
- ID or name columns may stay in the file; EasyClassifier recognises and leaves them out, and tells you.
- Old Excel files (.xls) must first be saved as .xlsx or .csv (in Excel: *File* → *Save As*).

6. Run it on your data - what each step means

Open the terminal in the folder where your data file is. Results are saved in a Results folder inside it.

- **Windows**: open the folder in File Explorer, click the address bar at the top, type cmd, press Enter.
- **macOS / Linux**: in Terminal, type cd (with a space), drag the folder from Finder/Files into the Terminal window, press Return.

Then start EasyClassifier (see the table in [section 4](#)). It shows *Step 1 of 6* to *Step 6 of 6*:

Step	What you see	What to do
Mode	Beginner / Advanced / Research	<i>Beginner</i> asks only essential questions; <i>Advanced</i> lets you review each preparation step.

Step	What you see	What to do
1. Load your data	"What is the name of your data file?" and the current folder	Type the file name (e.g. survey.xlsx), or drag the file into the window , then press ENTER.
2. Data inspection	Number of rows and columns, missing values, duplicates	Nothing – just read it.
3. Choose what to predict	Every column with a short description, e.g. Diagnosis - 2 categories (Healthy, Sick)	Press ENTER for the suggested column, or type its number. With many columns, type part of a name to search. If you pick an ID or a measurement, EasyClassifier explains and offers choices.
4. Prepare the data	What will be done with missing values, duplicates and text columns	Beginner: nothing. Advanced: choose each step (?N explains the options).
5. Choose classifier(s)	A list of methods	ENTER tries all of them (recommended). If you choose K-Nearest Neighbours, you also choose the distance – the Hassanat distance is suggested.
6. Review your choices	A summary	ENTER to start, or n to stop.

While it runs you see *Training ...* for each classifier. Then the results appear on screen, for example:
 Selected classifier: Support Vector Machine (SVM)

Final score on the untouched 20% test set - these are the numbers to report:

* Accuracy 96.67%

...

Comparison of all classifiers (balanced accuracy, ranked) - used only to choose the winner; slightly optimistic, do not report as the final result:

* Support Vector Machine (SVM) 96.60%

* Logistic Regression 96.60%

...

Which number should I report? The **final score** - not the best score in the comparison table. When several classifiers are compared, the winner partly won by luck; the final score is measured on data that played no part in choosing it (a 20 % test set kept aside from the start, or – for small datasets – nested cross-validation).

7. Your results

Every run creates a **new folder**, so earlier results are never overwritten:

```
Results/
survey_2026-09-28_10-15-02/
  report.pdf      ← start here
  report.tex
  summary.csv
  results.xlsx
  predictions.csv
  feature_importance.csv
```

```

trained_model.pkl
citations.txt
log.txt
figures/
  class_distribution.png
  comparison.png          (when several classifiers were compared)
  confusion_matrix.png
  roc_curve.png
  feature_importance.png
  ...                     (more, depending on your data and choices)

```

File	What it is
report.pdf	The full report in plain language: the question, the data, a Methods paragraph you can adapt for a paper , the results with an explanation of every number, the figures, which columns mattered, and the references to cite.
report.tex	The same report as a LaTeX file. Open it in Overleaf or any LaTeX editor to change the wording or layout.
summary.csv / results.xlsx	All scores in a table. The row starting with <i>FINAL</i> holds the numbers to report.
predictions.csv	For each tested row: the true class and the predicted class.
feature_importance.csv	How much each column helped the prediction.
trained_model.pkl	The trained model, including all preparation steps (for use by a programmer on new data).
citations.txt	What to cite if you publish.
log.txt	Every choice and step, with times – useful for your records and for reviewers.
figures/	The figures as images (300 dpi, ready for journals), explained in the report.

The figures

By default you get the five figures most used in classification papers. Two more are added automatically when your data need them. In *Advanced* mode you can pick any of them, and three optional extras.

Figure	When	What it shows
Class distribution	always	Rows in each class.
Classifier comparison	always, if several were compared	Each classifier's score in every test fold – is the winner clearly better? The report states this in words.
Confusion matrix	always	Which classes are confused, as counts and percentages.
ROC curves	always	How well each class is separated (one curve per class when there are 3 or more).
Feature importance	always	Which columns the model relies on.
Precision-recall curves	added when classes are unequal in size	The better view when one class is rare.

Figure	When	What it shows
Missing-value map	added when the data have empty cells	Where the gaps are.
Correlation matrix	optional	Which numeric columns move together.
Columns by class	optional	How the most important columns differ between classes.
Learning curve	optional	Would collecting more data help? The report answers in words. Needs at least 20 rows.

Colours and file formats (Advanced mode). Choose one of four themes: *colour-blind safe* (the default, recommended for papers), *greyscale* (for printed journals – classes are told apart by patterns and line styles), *high contrast* (slides and posters), or *soft*. A class keeps the same colour in every figure. Figures are saved as PNG at 300 dpi; you can also ask for PDF or SVG copies, which stay sharp at any size (PDF is best for journals; SVG can be edited in Inkscape or Illustrator).

8. How long does it take?

Data	Classifiers	Typical time
Up to a few hundred rows	all	about a minute
500 to 2,000 rows	all	a few minutes, sometimes more
500 to 2,000 rows	Random Forest, Logistic Regression, Naive Bayes, Decision Tree	about a minute
More than 2,000 rows	all	usually quicker than just below 2,000 (see below)
More than 20,000 rows	SVM, KNN or Neural Network	can take a long time; EasyClassifier warns you first

When several classifiers are compared, datasets of up to 2,000 rows use nested cross-validation for the final score: it repeats the whole comparison five times, which takes longer but gave the most accurate scores in our benchmarks. Larger datasets use a final test set, which is accurate there and much faster. Times depend on your computer.

9. Optional extras

PDF report (LaTeX)

report.pdf is made from report.tex by a LaTeX program. Without one, EasyClassifier still writes report.tex and tells you. To get PDFs:

- **Windows:** install [MiKTeX](#). When the installer asks what to do about missing packages, choose to install them automatically (*A/ways / Yes*). Otherwise MiKTeX opens a question window in the middle of making the report and the PDF step waits for it.
- **macOS:** install [MacTeX](#).
- **Linux (Ubuntu/Debian):** `sudo apt install texlive-latex-recommended texlive-latex-extra lmodern`

Or, without installing anything: make a zip file of the results folder, go to overleaf.com, choose *New project* → *Upload project*, upload the zip, and open `report.tex`. Overleaf shows the PDF.

Column names in non-Latin scripts (for example Arabic) appear as ? in the PDF. You can rename those columns in your data file, or edit `report.tex`.

More classifiers (XGBoost and LightGBM)

Windows	macOS
<code>py -m pip install "easyclassifier[full]"</code>	<code>python3 -m pip install "easyclassifier[full]"</code>

They then appear in the classifier list.

10. Updating and uninstalling

	Windows	macOS
Update	<code>py -m pip install --upgrade easyclassifier</code>	<code>python3 -m pip install --upgrade easyclassifier</code>
Uninstall	<code>py -m pip uninstall easyclassifier</code>	<code>python3 -m pip uninstall easyclassifier</code>

On Linux use `~/easyclassifier-env/bin/python -m pip ...`, or simply delete the `~/easyclassifier-env` folder to remove everything.

Your Results folders are never deleted by updating or uninstalling.

11. Troubleshooting

Message or problem	What to do
'easyclassifier' is not recognized... / command not found 'py' is not recognized (Windows)	Use <code>py -m easyclassifier</code> (Windows) or <code>python3 -m easyclassifier</code> (macOS). Python is not installed, or the classic installer was used without <i>Add python.exe to PATH</i> . Repeat section 2 .
python3: command not found (macOS) No module named easyclassifier	Install Python from python.org (section 2). It was installed for a different Python. Install and start it with the same first word (py, python3, or the environment path).
externally-managed-environment	Use the separate-environment steps in section 3 .
Installation fails with a very long file path (Windows)	Windows limits path length. Enable long paths (search the web for "Windows enable long paths", or ask your IT support) and install again.
The file cannot be found	Drag the file into the window instead of typing its name, or check the <i>Current folder</i> shown in Step 1.

Message or problem	What to do
A column of numbers is described as <i>categories</i> or <i>text values</i>	The column contains non-numbers such as units (72 kg) or notes (n/a). Clean the column in Excel and save again.
No LaTeX program was found	See section 9 ; <code>report.tex</code> has already been written.
LaTeX ... could not build the PDF	Details are in <code>report_latex_errors.log</code> . Uploading to Overleaf usually works.
It is very slow	Large data with SVM, KNN or Neural Network. Press Ctrl + C and choose faster classifiers (section 8).
Anything else	Every step is in <code>log.txt</code> . Please report problems on the project page, attaching <code>log.txt</code> (not your data).

12. Citing EasyClassifier

If you publish results obtained with EasyClassifier, please cite it. The exact references for your analysis are in `citations.txt` and at the end of `report.pdf`. They include the software, scikit-learn, NumPy, pandas and Matplotlib, a reference for every method that was used (each classifier, the KNN distance, the validation design, the measures and curves that need a source), and – when K-nearest neighbours with the Hassanat distance was used –

Hassanat, A. B., Alkafaween, E., Tarawneh, A. S., & Elmougy, S. (2022). Applications review of Hassanat distance metric. In *2022 International Conference on Emerging Trends in Computing and Engineering Applications (ETCEA)*, Karak, Jordan (pp. 1-6). IEEE. <https://doi.org/10.1109/ETCEA57049.2022.10009844>

Hassanat, A. B. (2014). Dimensionality invariant similarity measure. arXiv preprint arXiv:1409.0923. <https://arxiv.org/abs/1409.0923>

To cite the software itself: Hassanat, A. B. A. (2026). EasyClassifier (version 0.8.2). Zenodo. <https://doi.org/10.5281/zenodo.23122901>